

SHANGHAI PUDONG INTL, China

WMO: 583211

Lat: 31.143N

Lon: 121.805E

Elev: 13

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	30.2	8.8	8.6	37.2	14.0	11.2	37.8	25.6	37.1	23.2	39.7	8.0	300	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	95.2	81.1	92.9	80.9	89.9	79.9	82.9	90.2	81.9	88.4	81.0	87.1	11.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.9	160.6	86.8	80.4	158.1	86.3	79.1	151.1	85.2	46.7	91.1	45.6	89.7	44.8	87.4	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years					
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	21.7	19.7	DB	24.1	99.6	3.1	2.6	21.8	101.4	20.0	103.0	18.3	104.5	16.0	106.4
			WB	21.4	84.9	3.5	1.1	18.9	85.7	16.9	86.3	14.9	87.0	12.4	87.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.9	41.0	43.2	50.3	59.8	68.9	75.2	83.9	83.6	76.7	68.0	57.6	45.7
	DBStd	15.75	5.75	6.05	6.40	5.89	4.52	4.62	4.34	3.57	3.97	4.58	6.57	6.50
	HDD50	746	287	204	75	3	0	0	0	0	0	0	14	163
	HDD65	2857	745	610	457	176	15	0	0	0	0	22	234	598
	CDD50	5469	7	14	84	297	586	757	1052	1043	802	556	241	30
	CDD65	2105	0	0	1	21	135	307	587	578	352	113	11	0
	CDH74	19025	0	1	14	132	625	1803	6961	6793	2404	274	18	0
	CDH80	7047	0	0	2	27	140	528	3068	2783	477	22	0	0
Wind	WSAvg	10.6	10.6	10.8	10.9	10.8	10.6	9.7	10.5	11.1	10.9	10.6	10.0	10.4
Precipitation	PrecAvg	49.8	2.7	2.8	3.1	3.6	3.9	8.3	5.3	7.8	4.6	2.8	2.6	2.0
	PrecMax	66.4	7.9	6.3	7.3	6.3	8.7	27.0	9.5	15.8	10.5	12.5	5.2	4.6
	PrecMin	33.3	0.5	0.7	1.1	1.5	1.3	1.0	2.1	0.5	0.7	0.2	0.3	0.0
	PrecStd	8.9	1.7	1.5	1.6	1.3	1.8	5.9	2.2	3.9	2.2	3.0	1.6	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.3	66.5	74.8	84.0	88.1	93.6	99.0	98.5	89.9	82.6	76.6	66.1
		MCWB	53.6	58.3	61.9	66.2	70.0	80.1	81.1	81.9	78.3	75.4	66.4	59.3
	2%	DB	55.8	60.6	68.2	77.4	84.0	89.8	96.7	93.6	87.4	79.0	71.9	61.1
		MCWB	50.5	54.2	57.7	65.0	68.6	78.5	80.8	81.1	78.5	71.9	64.9	54.6
	5%	DB	52.1	55.6	64.4	73.3	80.5	85.9	93.6	91.5	84.4	77.0	69.8	57.6
		MCWB	47.8	50.0	55.2	62.9	68.1	76.9	81.0	80.9	76.8	69.3	64.0	52.7
	10%	DB	49.6	51.9	60.5	69.6	77.1	82.4	91.4	89.6	82.5	75.1	67.6	55.4
		MCWB	46.0	47.4	54.3	62.1	66.7	75.5	80.7	80.4	75.5	67.4	62.2	50.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.5	59.9	63.5	70.9	75.9	81.6	84.8	84.3	80.9	77.8	70.7	63.0
		MCDB	60.2	65.3	70.3	78.1	83.6	90.2	94.1	92.8	85.2	81.5	74.4	65.1
	2%	WB	52.1	55.2	60.1	67.4	72.9	79.8	83.3	82.9	79.7	74.7	68.5	57.0
		MCDB	54.6	59.5	65.6	72.9	77.9	87.4	91.5	89.6	84.1	77.7	71.1	60.2
	5%	WB	49.2	51.7	57.7	65.2	71.0	78.4	82.1	81.9	78.6	71.5	65.4	54.5
		MCDB	51.5	55.3	62.4	70.9	76.1	84.4	89.4	87.8	83.0	74.5	68.3	57.1
	10%	WB	46.0	48.7	55.1	62.9	69.3	76.7	81.1	81.2	77.4	69.1	63.2	51.5
		MCDB	48.5	50.8	59.8	68.5	74.1	81.4	88.0	86.8	81.6	73.4	66.2	54.7
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.1	13.4	14.0	13.1	10.0	11.6	10.0	9.2	10.9	12.2	12.3
		MCDBR	16.8	19.4	20.5	21.2	19.4	15.6	15.9	13.3	11.1	11.7	14.6	15.7
		MCWBR	11.2	12.7	11.8	11.0	8.9	6.9	5.6	5.2	5.4	6.5	8.8	10.6
	5% WB	MCDBR	13.8	15.9	17.7	17.0	14.2	13.7	14.0	11.5	10.0	9.5	10.9	12.4
	MCWBR	10.5	12.1	12.2	10.4	8.5	6.6	5.9	5.2	6.0	7.2	8.4	10.3	
Clear-Sky Solar Irradiance	taub		0.580	0.619	0.656	0.663	0.647	0.698	0.665	0.633	0.586	0.579	0.594	0.571
	taud		1.705	1.653	1.609	1.651	1.721	1.668	1.769	1.825	1.891	1.813	1.729	1.730
	Ebn at Noon		194	202	208	215	220	208	214	219	223	211	189	187
	Edh at Noon		63	72	81	80	75	79	71	66	60	61	61	58
All-Sky Solar Radiation	RadAvg		697	830	1179	1388	1503	1321	1703	1599	1289	1085	783	682
	RadStd		88	159	97	155	161	189	203	206	130	141	96	98

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	+106

Nomenclature: See separate page